

Question	Marking Guidance	Mark	Comments
09.1	1. LP due to mutation OR Allele due to mutation; 2. Milk provides named nutrient; 3. Individuals with LP more likely to survive and reproduce OR Individuals with advantageous <u>allele</u> more likely to survive and reproduce; 4. Directional selection; 5. Frequency of <u>allele</u> increases (in the offspring/next generation);	4 max	1. Reject mutation caused by drinking milk. 2. Accept any correct named nutrient e.g. glucose, galactose, protein 2. Ignore 'sugar' 'lactose' as named nutrient 1. and 3. Reject (LP) gene 3. Accept 'individuals who produce lactase' for 'LP individuals' 3. Accept 'pass on allele/LP/characteristic' for reproduce. 5. Accept description of increasing frequency of allele e.g. 'higher proportion', 'more common' but ignore increase in number of allele
09.2	1. Dominant <u>allele</u>; 2. (Always) expressed/shown (when present in phenotype/offspring) OR Expressed when only one (dominant allele) present;	2	
09.3	1. Mutation in promoter (DNA/gene) for transcription factor OR Mutation in promoter (region/DNA) for the gene OR Mutation in gene for transcription factor; 2. Lactase gene continues to be transcribed/active;	2	1. Accept mutation in an epistatic gene

